

Abstract

FitBuddy AI analyzes squat videos and delivers professional-grade feedback instantly: rep count, a 10-point form score, and corrective cues. Using **Google Gemini's multimodal reasoning**, five prompt strategies were benchmarked over 5 trials each and scored by an LLM judge against human-annotated ground truth. The **Self-Reflection Pass** scored highest (7.40/10), outperforming Zero-Shot (4.46), Three-Shot (5.81), and CoT-Verification alone (5.70). The production system combines Self-Reflection with CoT for maximum accuracy. Only 15 JPEG frames reach the Gemini API — raw video is never uploaded to any server.

Introduction & Motivation

Filming workouts is common — interpreting the footage is not. Without a coach, people cannot verify squat depth, knee tracking, or bracing, risking injury and stalled progress.

- No feedback loop:** recorded footage is useless without expert eyes
- Access gap:** personal trainers are expensive and unavailable
- Privacy:** existing tools require uploading full video to third-party servers

FitBuddy AI solves all three: instant rep count, form score, and corrective cues from a short video clip.

Materials & Methods

- Data collection:** Reddit r/formcheck + YouTube coaching breakdowns — videos paired with expert critique.
- Ground truth:** 40 videos standardized into a 4-file JSON hierarchy: *description / structured_feedback / evaluation*.
- Benchmark:** 5 prompt strategies × 5 trials, scored by Gemini 2.5 Flash (LLM judge) against human annotations.
- Deployment:** Best strategies combined. 15 frames via Canvas API → Gemini → JSON validation → dashboard.

Key Learnings

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Reflection is the highest-leverage intervention
Self-verification alone (7.40) outperformed Three-Shot (5.81) and CoT standalone (5.70).
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Measure before deciding
Without the evaluation harness, Three-Shot "felt" more robust — the judge ranked it 3rd.
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Client-side extraction removes infrastructure
15 JPEG frames only. Raw video never leaves the browser — no server costs, no privacy risk.
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Precision over recall for coaching
The reflection pass reduces false positives — the right tradeoff for a coaching tool.

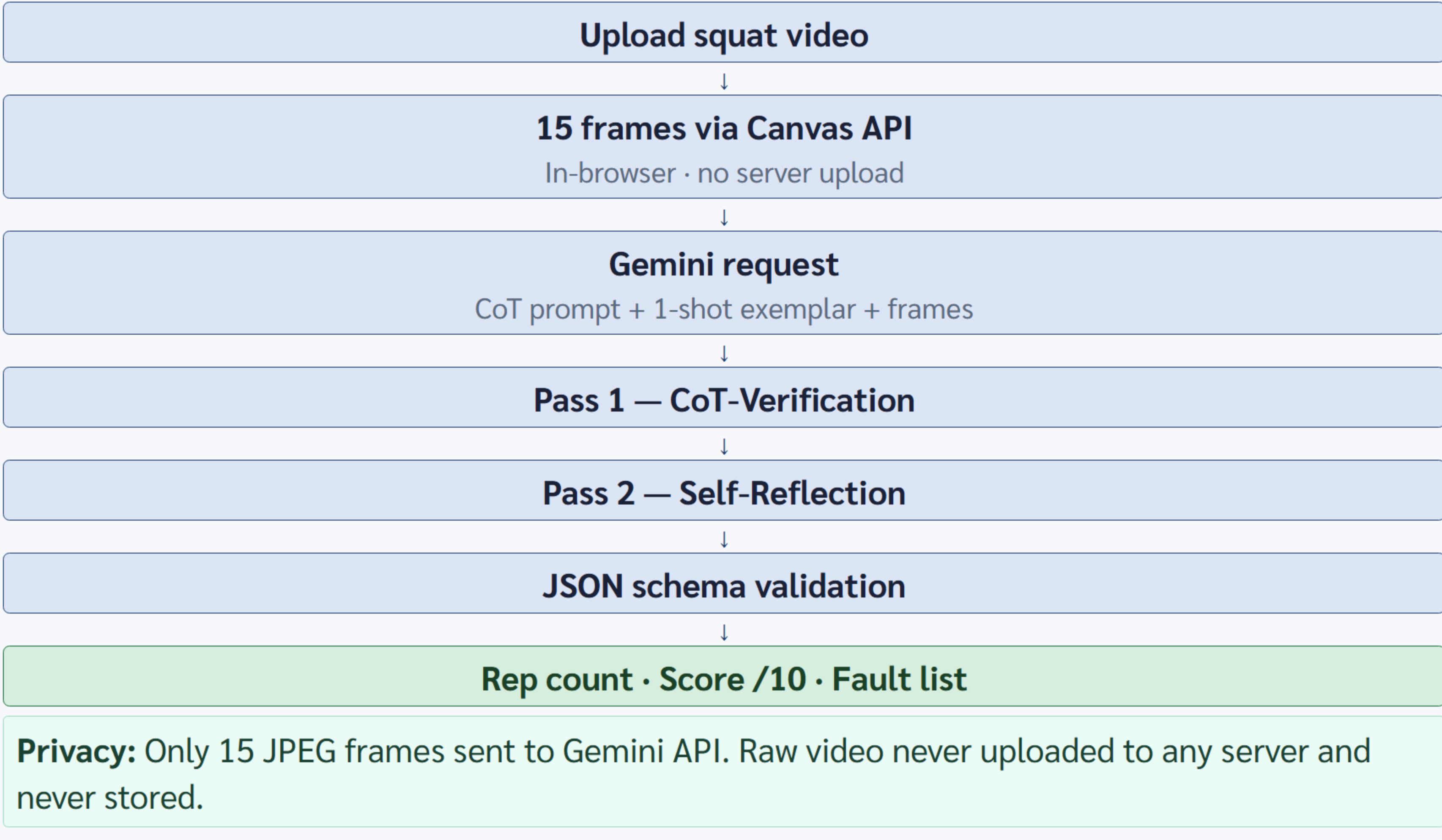
Future Improvements

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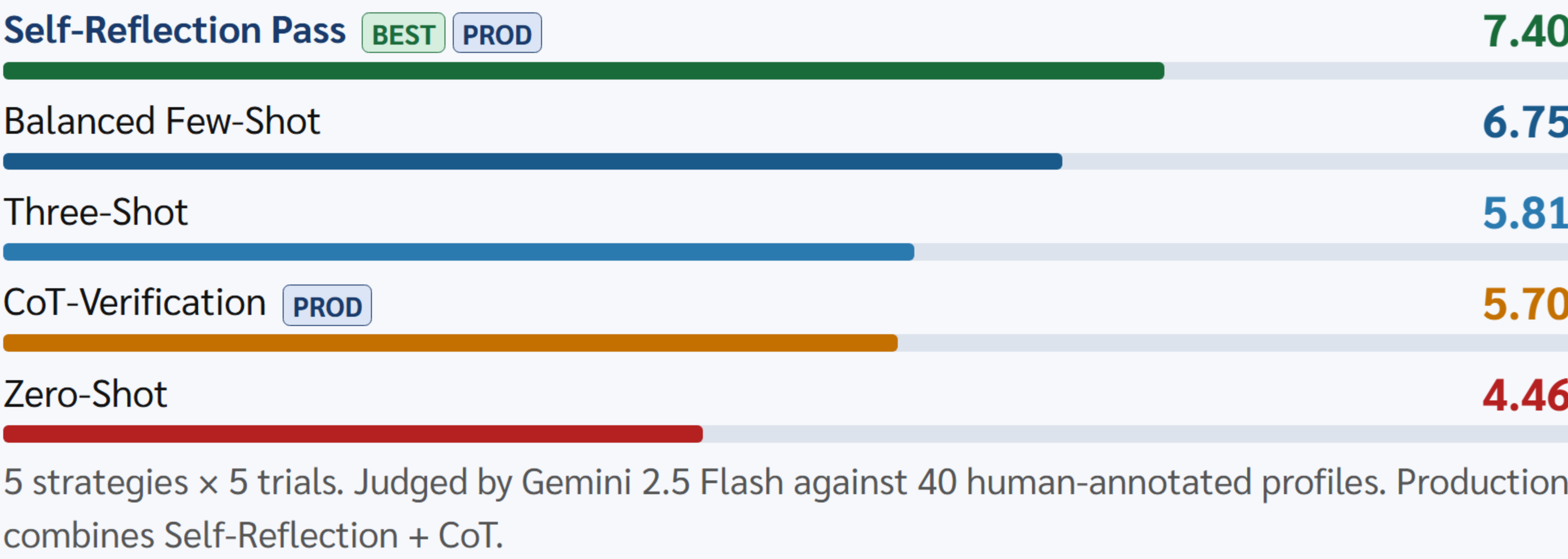
Rehab Mode Planned
Video-based rehabilitation tracking for recovery exercises outside the clinic.
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More Exercises Planned
Deadlift, push-up, bench press — same pipeline, same evaluation framework.

System Pipeline



Results · LLM Judge Score /10



Discussion & Conclusion

What this shows	The pattern generalizes	Broader impact
A self-verification pass — no extra training — raised alignment from 4.46 to 7.40/10 against human expert ground truth. The most impactful single change.	Sample frames in-browser → one verified exemplar → model self-verifies → enforce JSON schema. Applies to any video-based movement analysis task.	High-quality AI coaching without expensive infrastructure — expert-level feedback accessible to anyone with a smartphone and a short video.

References

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- [2] Zheng, L. et al. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *NeurIPS 2023*.
- [3] Google DeepMind. (2024). Gemini: A Family of Highly Capable Multimodal Models. *Technical Report*.
- [4] Reddit r/formcheck community. Squat form submissions used as human-annotated ground-truth training data.